30	α-particles are observed to be deflected by a thin metal foil through angles that range from 0° to 180°. Which of the following could explain this observation?	
	A	Due to diffraction from the crystal lattice in the metal.
	B	Due to reflection from the surface of the foil.
	C	Due to scattering from electrons in the metal.
	D	Due to scattering from the small highly positive charged nuclei of atoms in the metal.
L1	**Answer: D** Alpha particles are deflected by the nuclei of the metal atoms in the foil. These nuclei are small, heavy and positively charged. Since the scattering angle ranges to more than 90°, it cannot be attributed to particle exhibiting wave behaviour and diffracting.	